# AI-Based Medical Diagnosis Assistant - Diagnostic Report

Automated clinical decision-support report generated from ECG, EEG or chest radiograph input.

| | |
|---|---|
| **Analysis ID** | xray-20261029112618-51607b3a3d |
| **Modality** | XRAY |
| **Patient reference** | PT-0003 |
| **Age / Sex** | None / unknown |
| **Referring clinician** | - |
| **Generated at** | 2026-10-29T11:26:18.162068Z |
| **Model version** | 1.0.0 |
| **Model source** | trained |

## Primary Diagnosis

| | |
|---|---|
| **Condition** | **CONSOLIDATION** |
| **Confidence** | 100.0% |
| **Severity** | Moderate |
| **Interpretation** | Dense air-space consolidation suggesting infective or inflammatory change. |

## Class Probability Distribution

| Condition | Probability |
|---|---|
| CONSOLIDATION | 100.0% |
| NORMAL | 0.0% |
| PNEUMONIA | 0.0% |
| EFFUSION | 0.0% |
| TUBERCULOSIS | 0.0% |
| COVID19 | 0.0% |

## Clinical Findings

| Finding | Value | Reference | Status |
|---|---|---|---|
| Mean Intensity | 98.4 level | 20 - 120 level | Normal |
| Lung Opacity | 0.00 % | 0 - 25 % | Normal |
| Apical Opacity | 1.11 index | 0 - 0.6 index | Abnormal |
| Perihilar Opacity | 0.37 index | 0 - 1.3 index | Normal |
| Perihilar Peak | 0.95 index | 0 - 2 index | Normal |
| Peripheral Infiltration | 1.00 index | 0 - 0.55 index | Abnormal |
| Costophrenic Blunting | 0.80 index | 0 - 0.5 index | Abnormal |
| Symmetry Score | 0.95 index | 0.75 - 1 index | Normal |
| Texture Entropy | 5.47 bits | 4 - 7.2 bits | Normal |
| Edge Density | 34.9 % | 0.5 - 12 % | Abnormal |
| Image Quality | 0.17 index | 0.5 - 1 index | Abnormal |

## Explainable AI Analysis

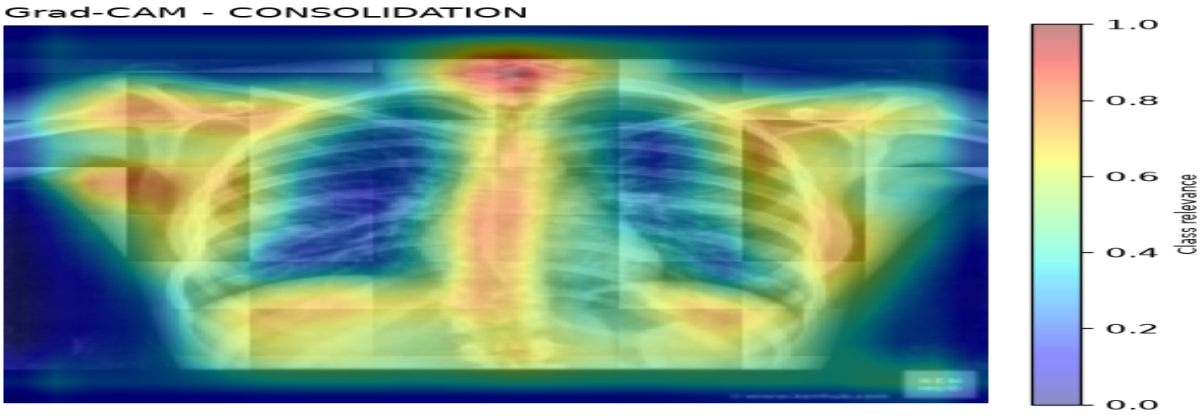
The prediction 'CONSOLIDATION' was driven mainly by Edge Density, Peripheral Infiltration, Texture Entropy. The model assigned 100.0% confidence to this class.

Attribution method: Grad-CAM over the final convolutional block

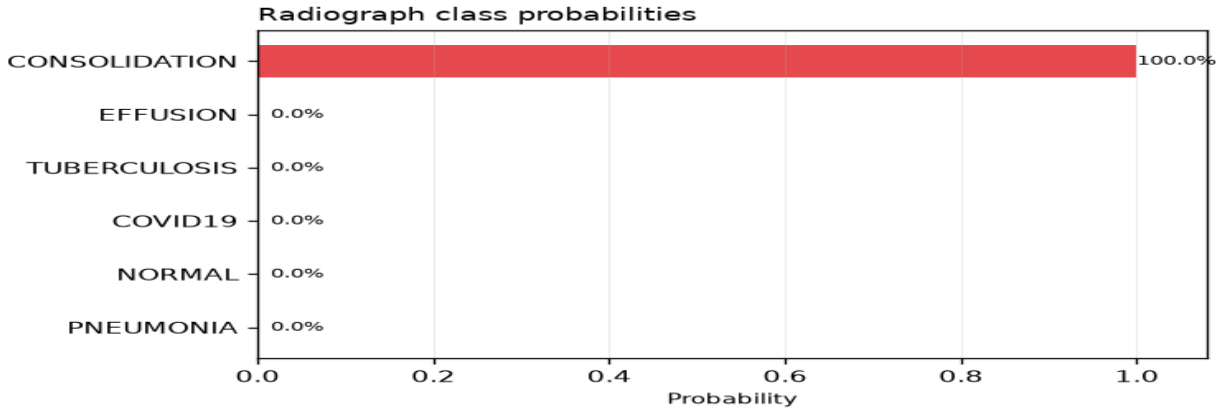
Signal attribution map




Grad-CAM localisation




Conditioned recording




## Leading Feature Contributions

| Feature | Contribution | Direction |
|---|---|---|
| Edge Density | 1.000 | Increases Risk |
| Peripheral Infiltration | 0.299 | Increases Risk |
| Texture Entropy | -0.298 | Decreases Risk |
| Perihilar Opacity | 0.251 | Increases Risk |
| Symmetry Score | -0.249 | Decreases Risk |
| Costophrenic Blunting | -0.113 | Decreases Risk |

| | | |
|---|---|---|
| Perihilar Peak | 0.080 | Increases Risk |
| Mean Intensity | 0.012 | Increases Risk |

## Clinical Recommendations

1. Air-space consolidation identified - evaluate for bacterial infection, aspiration or malignancy.
2. Compare with prior imaging to determine whether the finding is new.
3. A radiologist must confirm every finding reported by this decision-support tool.

## Input Quality

Signal quality index: **0.1684** | Usable: **no** | SNR: **0.0 dB**

This report was generated by an AI decision-support tool. It is not a medical diagnosis and must be interpreted by a qualified healthcare professional alongside the complete patient record, physical examination and confirmatory investigations.